Problem 2. Incomparable Pairs

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	512 mebibytes

You are given a string $s = s_1 s_2 \dots s_n$. Consider an unordered pair of its substrings $\{a, b\}$. Let us call such pair *incomparable* if neither a is a substring of b nor b is a substring of a. You have to compute the number of incomparable pairs of substrings of s.

Input

The first line of input contains a single string s consisting of lowercase English letters ($1 \le |s| \le 10^5$).

Output

Output a single integer which is the answer to the problem.

Examples

standard input	standard output
abba	8
abacaba	64

Problem 3. Aloha

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	256 mebibytes

Chiaki lives in a special universe where the the law of conservation of energy is quite different. Specifically, the law is described by the equation $v^2 + 2gy = 0$, where v is the object's speed at point (x, y) and g is the gravitational acceleration which equals to 10.

At first, Chiaki is at origin $(0, 0)$ without any kinetic energy, and she would like to go to point (x, y) using some pipes. Throughout her journey, the law of conservation of energy will apply to her.

Chiaki has three straight pipes with lengths l_1, l_2 and l_3. She must use the pipes to build a tunnel, which will be the route to the destination.

- At least one pipe should be used, each pipe can only be used once and the pipe cannot be bent or cut off.
- The tunnel must start from $(0, 0)$ and end at the destination (x, y).
- The pipes must be connected end to end.

Chiaki would like to know whether it is possible to reach the destination, and also the minimum time to go to the destination if it is possible.

Input

There are multiple test cases. The first line of the input contains an integer T ($1 \le T \le 500$), indicating the number of test cases. For each test case:

The first line contains five integers, x, y, l_1, l_2, and l_3 ($-1000 \le x, y \le 1000$, $1 \le l_1, l_2, l_3 \le 1000$): the coordinates of the destination and the length of each pipe.

Output

For each test case, output a single line containing a real number denoting the minimum time, or the string "`Impossible!`" (without the quotes) if Chiaki cannot reach the destination.

Your answer will be considered correct if and only if the absolute error or relative error of your answer is less than 10^{-8}.

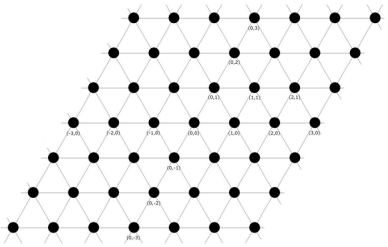
Example

standard input	standard output
4 0 1 1 1 1 3 -4 2 3 3 -1000 0 499 499 3 0 -8 10 2 1000	Impossible! 1.118033988749895 90.005448343698845 1.563516060678838

Problem 5. Triangular Lamps

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	512 mebibytes

There is an infinitely large triangular grid, as shown below. Each point with integer coordinates contains a lamp.



Initially, only the lamp at **(X, Y)** was on, and all other lamps were off. Then, Snuke performed the following operation zero or more times:

Choose two integers x and y. Toggle (on to off, off to on) the following three lamps: $(x, y), (x, y + 1), (x + 1, y)$.

After the operations, N lamps $(x_1, y_1), \cdots, (x_N, y_N)$are on, and all other lamps are off. Find X and Y.

Input

Input have the following format:

- Line 1: N
- Line 2: x_1 y_1
- ...
- Line $N + 1$: x_N y_N

Constraints:

$1 \leq N \leq 10^4$, $-10^{17} \leq x_i, y_i \leq 10^{17}$, (x_i, y_i) are pairwise distinct; the input is consistent with the statement, and you can uniquely determine X and Y.

Output

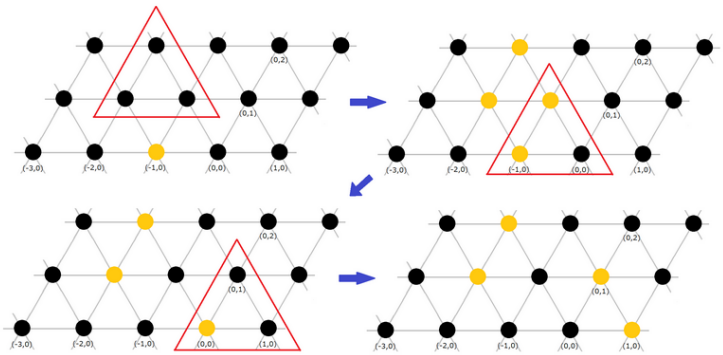
Print X and Y, separated by a space.

Example

standard input	standard output
4 -2 1 -2 2 0 1 1 0	-1 0

Note

The following picture shows one possible sequence of operations for the sample:


(0,2)
(0,1)
(-3,0)
(-2,0)
(-1,0)
(0,0)
(1,0)

Problem 7. Distinct Boxes

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Snuke has R red balls and B blue balls. He distributes them into K boxes, such that no box is empty and no two boxes are identical. Compute the maximum possible value of K.

Formally speaking, let's number the boxes 1 through K. If Box i contains r_i red balls and b_i blue balls, the following conditions must be satisfied:

- For each i $(1 \leq i \leq K)$, $r_i > 0$ or $b_i > 0$.
- For each i, j $(1 \leq i < j \leq K)$, $r_i \neq r_j$ or $b_i \neq b_j$.
- $\sum r_i = R$ and $\sum b_i = B$ (no balls can be left outside the boxes).

Input

Input is given in the following format:

- Line 1: R B

Constraints:

$1 \leq R, B \leq 10^9$

Output

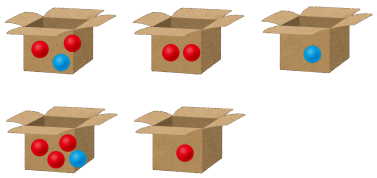
Print the maximum possible value of K.

Example

standard input	standard output
8 3	5

Note

The following picture shows one possible way to achieve $K = 5$:



Problem 11. e

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

There is a very long bench. The bench is divided into M sections, where M is a very large integer.

Initially, the bench is vacant. Then, M people come to the bench one by one, and perform the following action:

We call a section *comfortable* if the section is currently unoccupied and is not adjacent to any occupied sections. If there is no comfortable section, the person leaves the bench. Otherwise, the person chooses one of comfortable sections uniformly at random, and sits there. (The choices are independent from each other).

After all M people perform actions, Snuke chooses an interval of N consecutive sections uniformly at random (from $M - N + 1$ possible intervals), and takes a photo. His photo can be described by a string of length N consisting of '`X`' and '`-`': the i-th character of the string is '`X`' if the i-th section from the left in the interval is occupied, and '`-`' otherwise. Note that the photo is directed. For example, "`-X--X`" and "`X--X-`" are different photos.

What is the probability that the photo matches a given string s? This probability depends on M. You need to compute the limit of this probability when M goes infinity.

Here, we can prove that the limit can be uniquely written in the following format using three *rational* numbers p, q, r and $e = 2.718\ldots$ (the base of natural logarithm):

$$p + \frac{q}{e} + \frac{r}{e^2}$$

Your task is to compute these three rational numbers, and print them modulo $10^9 + 7$, as described in Output format.

Input

Input is given in the following format:

- Line 1: N
- Line 2: s

Constraints:

- $1 \leq N \leq 1000$
- $|s| = N$
- s consists of '`X` and '`-`'.

Output

Print three rational numbers p, q, r, separated by spaces. When you print a rational number, first write it as a fraction $\frac{y}{x}$, where x, y are integers and x is not divisible by $10^9 + 7$ (under the constraints of the problem, such representation is always possible). Then, you need to print the only integer z between 0 and $10^9 + 6$, inclusive, that satisfies $xz \equiv y \pmod{10^9 + 7}$.

Examples

standard input	standard output
1 X	500000004 0 500000003
3 ---	0 0 0
5 X--X-	0 0 1
5 X-X-X	500000004 0 833333337
20 -X--X--X-X--X--X-X-X	0 0 183703705

Note

For Sample 1, the probability that a randomly chosen section is occupied converge to $\frac{1}{2} - \frac{1}{2e^2}$.

For Sample 2, after the actions, no three consecutive unoccupied sections can be left.

For Sample 3, the limit is $\frac{1}{e^2}$.

For Sample 4, the limit is $\frac{1}{2} - \frac{13}{6e^2}$.

For Sample 5, the limit is $\frac{7}{675e^2}$.

The `samples.zip` file contains six samples; in sixth sample $N = 100$.

Problem 13. Game

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

There are two arrays of integers a and b of length n and m, respectively.

There is a chosen position in each array. Denote the chosen position in a as c and in b as d. The **state** is a tuple (c, d). The chosen positions are never out of bounds, i.e. $1 \le c \le n$ and $1 \le d \le m$. At the start of the game both c and d are equal to 1.

Two players are playing a game taking turns alternately. On his turn each player must do one of the following:

1. Move. Change the chosen position in one of the arrays. You can't change a chosen position to itself. It is not allowed to make a move which will result in a state appearing for the 10^{228}-th time. The starting state is considered as having appeared once at the beginning of the game.
2. *Screw you guys, I'm going home.* Finish the game.

Note that there are $n \cdot m$ states and each state appears no more than $10^{228} - 1$ times. Therefore the game is finite.

The score of the game is equal to the sum of elements on the chosen positions, i.e. $a_c + b_d$. The player who moves first minimizes it by playing accordingly and the one who moves second maximizes it.

What is the score of the game assuming perfect play from both sides?

Input

The first line contains two n and m ($1 \le n, m \le 10^5$), lengths of arrays a and b, respectively.

The second line contains n integers a_i ($1 \le a_i \le 10^8$), the elements of a.

The third line contains m integers b_i ($1 \le b_i \le 10^8$), the elements of b.

Output

Output a single integer — the score of the game.

Examples

standard input	standard output
2 2 1 3 1 3	2
2 1 3 2 2	4
3 1 3 3 2 2	5
4 5 10 1 2 3 10 1 2 3 4	11
6 7 4 5 6 1 2 3 5 2 1 3 4 6 7	7
4 3 64 16 4 1 32 8 2	33

Problem 17. Three Dimensions

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	256 mebibytes

Let's define a strange "distance" between two points $a = (x_a, y_a, z_a)$ and $b = (x_b, y_b, z_b)$ in three-dimensional space:

$$d(a,b) = \max\{|x_a - x_b|, |y_a - y_b|, |z_a - z_b|\} \oplus x_a \oplus y_a \oplus z_a \oplus x_b \oplus y_b \oplus z_b,$$

where $\max\{S\}$, $|x|$ and $\oplus$ correspond to the maximum value in S, the absolute value of x and the bitwise exclusive-or operator respectively.

Given 6 non-negative integers $mx_a, my_a, mz_a, mx_b, my_b, mz_b$, please find the sum of all $d(a,b)$, where $x_a \in [0, mx_a], y_a \in [0, my_a], z_a \in [0, mz_a]$ and $x_b \in [0, mx_b], y_b \in [0, my_b], z_b \in [0, mz_b]$. Since the sum may be very large, please output it modulo 2^{30} instead.

Note that $x_a, y_a, z_a, x_b, y_b, z_b$ should all be integers.

Input

The input only contains 6 non-negative integers $mx_a, my_a, mz_a, mx_b, my_b, mz_b$, each integer is not larger than 10^9.

Output

Output an integer denoting the sum modulo 2^{30}.

Example

standard input	standard output
3 2 1 2 1 3	778

Problem 19. Parity Scam

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

This is an interactive problem.

Pavel is a famous prosecutor. Throughout his career, he had convicted a lot of scammers, money-launderers and other criminals who plagued the city of Baǵel. Recently a major scam scheme was uncovered. The scammers had a connected undirected graph $G = \langle V, E \rangle$ with maximum vertex degree **at most** 10, and a function $f : V \to \{0, 1\}$. They have suggested people to try find a function $g : E \to \{0, 1\}$ such that, for every $v \in V$, the equality $\bigoplus_{e \in I(v)} g(e) = f(v)$ holds. Here, $I(v)$ is the set of edges of G incident to v, and $\oplus$ is sum modulo 2.

Pavel has quickly noticed that $\bigoplus_{v \in V} f(v) = 1$ and concluded that such g does not exist! But how to convince the jury that it is so?

Sam, the leader of this scam group, claims that such g exists and that he knows exactly what it is. Pavel is going to ask him questions about this claimed g until he finds a *simple contradiction* in Sam's answers.

Pavel can only ask questions in the form of boolean formulas with variables corresponding to the values of g. Let us enumerate the edges and identify them with their numbers. Let us define a boolean formula:

- `g`x, where $x \in \{1, 2, \ldots, |E|\}$, is a boolean formula.
- If F is a boolean formula, then `!(`F`)` is a boolean formula.
- If F and G are boolean formulas, then `(`F`)&(`G`)` and `(`F`)|(`G`)` are boolean formulas.

Sam answers 0 (false) or 1 (true) to each question. The types of *simple contradictions* that the jury recognizes are the following:

1. The answer to F is α, and the answer to `!(`F`)` is also α.
2. The answer to F is α, the answer to G is β, and the answer to `(`F`)&(`G`)` is not $\alpha \wedge \beta$ (here, $\wedge$ is logical and).
3. The answer to F is α, the answer to G is β, and the answer to `(`F`)|(`G`)` is not $\alpha \vee \beta$ (here, $\vee$ is logical or).

Formally, Sam's claim that g satisfies the parity condition for every $v \in V$ which means that, for every $v \in V$, the equality $\bigoplus_{e \in I(v)} g(e) = f(v)$ holds, is interpreted as follows. For each vertex $v \in V$ and each subset $S \subseteq I(v)$ such that its size $|S| \bmod 2 \neq f(v)$, it is assumed that Sam has answered 1 to the formula

$$\bigvee_{i \in S} \texttt{!(g}i\texttt{)} \vee \bigvee_{i \in I(v) \setminus S} \texttt{g}i.$$

We refer to these formulas as **axioms**. Pavel can ask them, and Sam will answer 1.

Here we assume that the whole big disjunction is written in right-associative form. For example, the formula $x \vee y \vee z$ is written as `(x)|((y)|(z))`. The variables in the axioms are ordered so that their numbers are increasing from left to right. So, for example, an axiom $\bigvee_{i \in \{1,3\}} \texttt{!(g}i\texttt{)} \vee \bigvee_{i \in \{2\}} \texttt{g}i$ is written as `(!(g1))|((g2)|(!(g3)))`.

Pavel would like to convince the jury as soon as possible, so he would like to ask at most 50 questions.

Additionally, Pavel would like to ask formulas that are not too deep, so the jury wouldn't become confused. The depth $d(F)$ of a formula F is defined recursively:

- $d(\texttt{g}i) = 1$ for $i \in \{1, \ldots, |E|\}$.
- $d(\texttt{!(}F\texttt{)}) = d(F) + 1$.
- $d(\texttt{(}F\texttt{)|(}G\texttt{)}) = \max\{d(F), d(G)\} + 1$.
- $d(\texttt{(}F\texttt{)\&(}G\texttt{)}) = \max\{d(F), d(G)\} + 1$.

Every formula that Pavel asks must have depth at most 100.

Input

The first line contains two integers n and m ($1 \le n \le 100$, $1 \le m \le 150$): the number of vertices and the number of edges in G respectively. The vertices are numbered from 1 to n, and the edges are numbered from 1 to m.

The second line contains n integers $f(1), f(2), \ldots, f(n) \in \{0, 1\}$. It is guaranteed that their sum is odd.

The next m lines contain the description of the edges of G. Each of these lines contains two distinct integers from the set $\{1, \ldots, n\}$: the ends of the corresponding edge.

It is guaranteed that the given graph is connected and has maximum vertex degree at most 10. Note that the graph may contain parallel edges.

Interaction Protocol

As soon as it reads the input, your solution may ask four types of queries, each on a separate line:

1. "`?` F" where F is a boolean formula in variables `g1`, ..., `g`n. The jury then prints Sam's answer on a separate line. **If F is an axiom, written in right-associative form exactly as described above, it is guaranteed that Sam's answer will be 1.**
2. "`!` i_1 i_2" where i_1, i_2 are numbers of questions F and G such that $G =$ `!(`F`)` and they form a simple contradiction of the first type.
3. "`|` i_1 i_2 i_3" where i_1, i_2, i_3 are numbers of questions F, G, and H such that $H =$ `(`F`)|(`G`)` and they form a simple contradiction of the second type.
4. "`&` i_1 i_2 i_3" where i_1, i_2, i_3 are numbers of questions F, G, and H such that $H =$ `(`F`)&(`G`)` and they form a simple contradiction of the third type.

After a query of type 2, 3 or 4, your solution must immediately terminate gracefully. Your solution must not make more than 50 queries of the first type, and each such query must have depth at most 100. The total length of all formulas should not exceed 2 000 000 characters.

Note that Sam can answer any other questions than the axioms in any way he likes. He may even answer differently to the same repeated question!

Do not forget to flush the output buffer after printing each line! It can be done explicitly by calling, for example, `fflush(stdout)` in C or C++, `System.out.flush()` in Java, `flush(output)` in Pascal, or `sys.stdout.flush()` in Python.

Example

standard input	standard output
3 3 1 1 1 1 2 2 3 1 3 1 1 1 0 0	 ? g1 ? g2 ? (!(g1))|(!(g2)) ? !(g1) ? !(g2) | 4 5 3

Example explanation

In the example, the boolean formula "`(!(g1))|(!(g2))`" is an axiom.

Problem 23. Beautiful Automata

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

Oleksandr is a fan of strings. His favourite data structure is suffix automaton.

Consider a string s of lowercase English letters. The suffix automaton of s is the smallest (having the smallest number of vertices) directed acyclic graph G with a letter $l(e)$ written on each edge e and a fixed starting vertex v_0 such that

$$\{w \mid w \text{ is a substring of } s\} = \{l(e_1)l(e_2)...l(e_k) \mid (e_1, e_2, ..., e_k) \text{ is a path in } G \text{ starting at } v_0\}.$$

Oleksandr likes suffix automata more than any other graphs. He calls a directed acyclic graph G *s-beautiful* if it is possible to write a lowercase English letter on each edge and choose a starting vertex v_0 so that G will become a suffix automaton of the string s. Oleksandr likes lexicographically small strings, so please help him find for a given graph G the lexicographically smallest string s such that G is s-beautiful.

Input

The first line contains two integers n and m ($1 \le n \le 2000$, $1 \le m \le 3000$), the number of vertices and the number of edges in G respectively. Each of the next m lines contains two integers v and u ($1 \le v, u \le n$, $v \ne u$), denoting a directed edge from v to u. It is guaranteed that G is acyclic.

Output

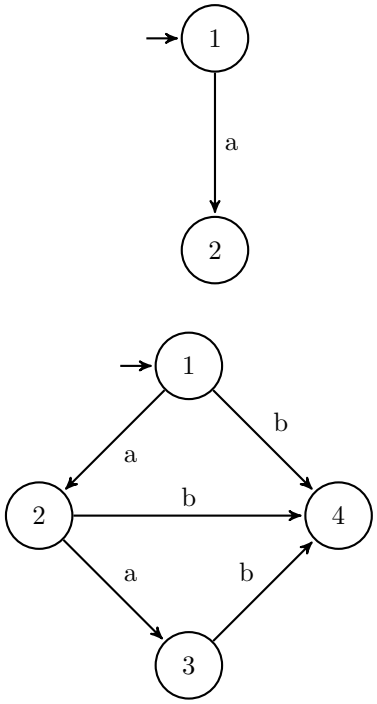
Output a single line containing the lexicographically smallest string s consisting of lowercase English letters such that G is s-beautiful. If such string does not exist, output -1.

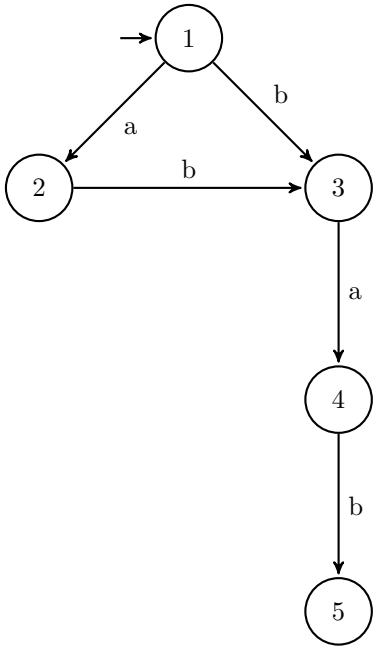
Examples

standard input	standard output
2 1 1 2	a
4 5 1 2 2 3 1 4 2 4 3 4	aab
5 5 1 2 1 3 2 3 3 4 4 5	abab
4 5 1 2 1 3 1 4 2 3 4 3	-1

Explanations

Suffix automata for the first three sample tests are shown below:





Problem 29. Hacked "Lines"

Input file:	*standard input*
Output file:	*standard output*
Time limit:	3 seconds
Memory limit:	256 mebibytes

In this task, we will tell about the classic version of the game "Lines". There is a square field of 9×9 cells. Each cell can be empty or contain a single ball of one of 7 colors. Each turn the computer puts three balls of random colors in random empty cells. After that player can move one ball in any empty cell. If 5 or more balls of the same color arranged in a vertical , horizontal or diagonal lines, these balls are destroyed, and the player gets points and an extra turn.

In this task, everything is a bit different: we hacked the computer and are now able to select empty cells on each turn, which will be occupied by new balls, and colors of these new balls. Two new balls appear on each turn. Player doesn't move anything (because we are directing the actions of the computer). Our goal — to make such moves, that allow us on the first destruction to destroy as many balls as possible. That is, you can make a few moves to build some successful design, and then destroy it in one move . You cannot make additional destructions to clear map.

Two balls are put in the field simultaneously. Thereafter, any ball that belongs to the vertical, horizontal or diagonal line of at least 5 balls of the same color, instantly collapses. If there are less than two empty cells, the next turn is not available.

Input

Input contains exactly 9 rows by 9 characters each — the current position in the game. Each character is either a digit from 1 to 7 , meaning the color of the ball in the current cell, or character '.' if the cell is empty. It is guaranteed that there is no line with 5 or more balls of the same color.

Output

The first line of output should contain two numbers D and M — maximum number of balls that can be destroyed by one stroke and the number of moves that lead to this destruction. $2M$ lines follow(2 lines on each turn). Each line should contain three numbers r_i, c_i and $color_i$ ($1 \le r_i, c_i \le 9$ $1 \le color_i \le 7$) — number of row and column of empty cells, where ball of $color_i$ color was placed. Both ball of one turn were placed at the same time. At the time of first destruction you must destroy exactly D balls (theoretically its possible to destroy them in a few times and make extra moves, but this is not recommended).

If no one ball can be destroyed, D must be 0. M does not exceed 100.

Examples

standard input	standard output
723134552 2421.2456 353...442 14...2..4 23.2....4 4....1..1 1222..112 3621..124 631211777	25 9 6 7 2 3 4 2 5 8 2 5 7 2 5 3 2 6 3 2 3 6 2 8 5 2 7 5 2 6 5 2 5 5 2 2 5 2 3 5 2 4 8 2 4 7 2 4 4 2 5 6 2 4 5 2

Problem 31. Bohemian Rhaksody

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	1024 mebibytes

Is this the real life? Is this just fantasy?
Caught in a physics formula, no escape from reality...

MolaMola is a world-famous rock band composed of three members: Guitarist Sangsoo `cki86201` Park famous for his amazing performance with Link-cut tree, drummer Sunggwan `dotorya` Park who plays 5 problems in 10 minutes without single WA, and the vocalist Bumsoo `zlzmsrhak` Park who is responsible for the impressive outlook of the band. The movie about their story, *Bohemian Rhaksody*, became a blockbuster in Korea, and they are preparing for their perfect comeback concert.

Bumsoo, famous for his unique sense of fashion, thinks that perfect concerts need not only the best music but also great visual effects. Needless to say, the illumination setting is a very important matter for him.

The stage given to them is a $H \times W$ size rectangle with four corners in $(0,0), (0,W), (H,0), (H,W)$, and there are N bulbs already installed in the stage. Note that the bulbs may lie on the boundary, but are not strictly outside. To prevent interference, each bulb has a distinct x-coordinate, and each bulb also has a distinct y-coordinate. Every bulb is designed to light either the east/west/south/north part of the stage. More formally, the bulb at the coordinate (X,Y) can be operated in one of the following modes:

- Light the area of the stage where $x \leq X$.
- Light the area of the stage where $x \geq X$.
- Light the area of the stage where $y \leq Y$.
- Light the area of the stage where $y \geq Y$.

Bumsoo wants their band to play in the brightest area, so the stage will be exactly the area that is lit by all N bulbs. Now, Bumsoo wants to find a way to operate the bulbs, in order to maximize the area of their stage. Help him!

Input

The first line contains three integers H, W, N. $(1 \leq N \leq 100000, 1 \leq H, W \leq 10^8)$

In the next N lines, two integers X, Y are given, indicating that the bulb is installed in coordinate (X,Y). $(0 \leq X \leq H, 0 \leq Y \leq W)$

It is guaranteed that no pair of different bulbs share an x-coordinate, and no pair of different bulbs share a y-coordinate.

Output

Print the maximum possible area that can be lit by all N bulbs.

Examples

standard input	standard output
4 4 5 0 4 1 3 2 2 3 1 4 0	6
100000000 100000000 1 0 0	10000000000000000
100000000 100000000 12 100000000 59411855 0 4914151 57454627 45388814 93661922 93279520 81531691 0 5221549 64790529 75886863 85609174 74950464 100000000 18493301 57818271 66752434 90450964 44757377 54518291 99631520 21997156	4522156529817280

Problem 37. Good Permutations

Input file:	*standard input*
Output file:	*standard output*
Time limit:	7 seconds
Memory limit:	256 mebibytes

Let's call a permutation of n elements **good**, if there are exactly m triples i, j, k such that $1 \le i < j < k \le n$ and $p_i < p_j < p_k$.

You need to calculate the total number of inversions of all good permutations of n elements, modulo 998 244 353 (prime).

Input

The first line of input contains two integers n and m ($1 \le n \le 100\,000, 0 \le m \le 3$).

Output

Output one integer: the sum of the number of inversions of all permutations $p_1, p_2, \ldots, p_n$, such that there are exactly m triples i, j, k such that $1 \le i < j < k \le n$ and $p_i < p_j < p_k$, modulo 998 244 353.

Examples

standard input	standard output
2 0	1
3 0	9
4 0	55
5 0	290
4 2	3
5 2	98
6 2	1074
5 3	21
6 3	484
7 3	5430

Problem 41. Elections

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	256 mebibytes

An election selecting the members of the parliament in Byteland was held. The only system adopted in this country is the party-list proportional representation. In this system, each citizen votes for a political party, and the number of seats a party wins will be proportional to the number of votes it receives. Since the total number of seats in the parliament is an integer, of course, it is often impossible to allocate seats exactly proportionaly. In Byteland, the following method, known as the D'Hondt method, is used to determine the number of seats for each party.

Assume that every party has an unlimited supply of candidates and the candidates of each party are ordered in some way. To the y-th candidate of a party which received x votes, assign the value x/y. Then all the candidates are sorted in the decreasing order of their assigned values. The first T candidates are considered to win, where T is the total number of seats, and the number of seats a party win is the number of its winning candidates.

The table below shows an example with three parties. The first party received 40 votes, the second 60 votes, and the third 30 votes. If the total number of seats is $T = 9$, the first party will win 3 seats, the second 4 seats, and the third 2 seats.

# of votes	x/1	x/2	x/3	x/4	x/5	x/6	x/
40	40	20	13.33...	10	8	6.66...	5.7
60	60	30	20	15	12	10	6.57
30	30	15	10	7.5	6	5	3.28

When selecting winning candidates, ties are broken by lottery; any tied candidates will have a chance to win. For instance, in the example above, if $T = 5$ then two candidates tie for the value 20 and there are two possible outcomes:

- The first party wins 2 seats, the second 2 seats, and the third 1 seat.
- The first party wins 1 seat, the second 3 seats, and the third 1 seat.

You have just heard the results of the election on TV. Knowing the total number of valid votes and the number of seats each party won, you wonder how many votes each party received.

Given N, M, and S_i ($1 \le i \le M$), denoting the total number of valid votes, the number of parties, and the number of seats the i-th party won, respectively, your task is to determine for each party the minimum and the maximum possible number of votes it received. Note that for some cases there might be no such situation with the given N, M, and S_i.

Input

The first line of the input contains two integers N ($1 \le N \le 10^9$) and M ($1 \le M \le 3 \cdot 10^4$), where N is the total number of valid votes and M is the number of parties. M lines follow, the i-th of which contains a single integer S_i ($0 \le S_i \le 3 \cdot 10^4$), representing the number of seats the i-th party won. You can assume that there exists i with $S_i \neq 0$.

Output

If there is no situation with the given N, M, and S_i, display the word "`impossible`". Otherwise, output M lines, each containing two integers. The first integer and the second integer in the i-th line should be the minimum and the maximum possible number of votes the i-th party received, respectively.

Examples

standard input	standard output
10 2 2 1	5 7 3 5
5 6 0 2 0 2 6 0	0 0 1 1 0 0 1 1 3 3 0 0
2000 5 200 201 202 203 204	396 397 397 399 399 401 401 403 403 404
15 10 1 1 1 1 1 1 1 1 1 13	impossible
1000000000 9 12507 16653 26746 21516 29090 10215 28375 21379 18494	67611619 67619582 90024490 90033301 144586260 144597136 116313392 116323198 157257695 157269050 55221291 55228786 153392475 153403684 115572783 115582561 99976756 99985943

Problem 43. Zayin and Fireball

Input file:	*standard input*
Output file:	*standard output*
Time limit:	20 seconds
Memory limit:	512 mebibytes

Zayin is practicing magic at home. His house is an infinite 2-dimension plane. He can summon a fireball which burns a circle area to the ground. To prevent the house from lying in ruins, his dad Nothingnuo has to cast shield magic. The shield can protect a circle area from being destroyed.

This game repeat for N times. Each time Zayin summons a fireball and Nothingnuo places a shield. After the exploding, this shield will no longer work in the future. The process leaves some (or no) area ruined. And it will never recover.

Nothing wonders about the union of the ruined area.

Zayin's i-th fireball destroyed a circle area. It has its center located at (X_i, Y_i) and has a radius R_i. Nothing's shields are described similarly. For more details, you can refer to **Input**.

In short: There are $2N$ circles. Which can be described as $\{X_1, Y_1, R_1\}, \{X_2, Y_2, R_2\} \dots \{X_N, Y_N, R_N\}$ and $\{x_1, y_1, r_1\}, \{x_2, y_2, r_2\} \dots \{x_N, y_N, r_N\}$. (X, Y) is the coordinate of center, and R is radius. And you need to calculate $AreaSize(\cup_1^N \{X_i, Y_i, R_i\} \setminus \{x_i, y_i, r_i\})$.

Input

The first line of input contains an integer $T(1 \le T \le 30)$, denoting the number of test cases.

Each test case starts with a positive integer $N(1 \le N \le 500)$.

The next N lines give the description of the circles. The i-th line contains 6 integers: $X_i, Y_i, R_i, x_i, y_i, r_i$. $(|X|, |Y|, |x|, |y| \le 1000, 0 \le R, r \le 200)$.

Notice: It is guaranteed that all $\{X_i, Y_i, R_i\}$ are different with each other. And all $\{x_i, y_i, r_i\}$ are different with each other. But there may exist i, j that $\{X_i, Y_i, R_i\}$ and $\{x_j, y_j, r_j\}$ are the same.

Output

For each case print the ruined area.

Absolute error up to 10^{-2} will be ignored.

Example

standard input	standard output
2 1 0 0 2 0 0 1 2 0 0 2 2 0 2 2 2 2 233 0 51	9.424778 19.707963

Problem 47. Zayin and Dirichlet

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	512 mebibytes

We define three types of **fundamental functions**,

$$\mu(n) = \begin{cases} (-1)^\sigma & , n \text{ has no square factor,} \\ & \quad \textit{and } \sigma \text{ is the number of factors of } n \\ 0 & , \text{otherwise} \end{cases}$$

$$1(n) = 1$$

$$id_k(n) = n^k \qquad , k = 1, 2, 3, \ldots$$

The Dirichlet product of functions f and g (represented as $f * g$) is a function h where

$$h(n) = \sum_{d|n} f(d)g(\frac{n}{d})$$

The Dirichlet product of more than two functions (for example, there are q functions $f_1, f_2, \cdots, f_q$) is defined as

$$f = (((f_1 * f_2) * f_3) * \cdots) * f_q$$

Zayin and Ziyin now have many **fundamental functions** (at least one). They calculate the Dirichlet product of them all and get the result f. Zayin finds that for any prime p, $f(p^c)$ is a polynomial of p of degree n, i.e. $f(p^c) = \sum_{i=0}^{n} a_i p^i$. But a_i is so large that he can only tell you a_i mod 998244353, i.e. x_i where $0 \le x_i < 998244353$ and $x_i \equiv a_i \pmod{998244353}$ holds.

And Ziyin finds that for any prime p, $f(p^d)$ is also a polynomial of p, but she doesn't want to tell you what it looks like. So it's your task to find out that polynomial.

If there are multiple solutions, please print the smallest one. Polynomial $P(x) = \sum_{i=0}^{n} a_i x^i$ is smaller than $Q(x) = \sum_{i=0}^{m} b_i x^i$ if and only if $n < m$, or $n = m$ and $\exists i \in [0, n],\ (\forall j \in [i+1, n]\ a_j = b_j) \wedge (a_i < b_i)$ (after modulo 998244353).

If there is no solution, or the number of fundamental functions used by the smallest solution is more than 10^5, please print -1.

Input

The first line contains three integers n, c and d. ($0 \le n \le 1000$, $1 \le d \le c \le 100$)

The second line contains $n+1$ integers $x_0, x_1, \ldots, x_n$. ($0 \le x_i < 998244353,\ x_n \ne 0$)

Output

Print -1 in the only line if it satisfies the corresponding conditions above. Otherwise the output contains two lines.

The first line contains one integer m, representing the degree of Ziyin's polynomial.

The second line contains $m+1$ integers $b_0, b_1, \cdots, b_m$, representing $f(p^d) = \sum_{i=0}^{m} b_i p^i$. You must make sure $b_m \ne 0$ when $m > 0$. Due to the answers are so large, please print answers modulo 998244353.

Example

standard input	standard output
2 2 1 1 1 1	1 1 1
2 2 1 998244352 998244352 1	-1

Note

In the first sample, $f(p^2) = p^2 + p + 1$. $f(n)$ can be regarded as the sum of divisors of n, so that $f(p) = p + 1$.

In the second sample, $f(p^2) = p^2 - p - 1$ can not be Dirichlet product of fundamental functions.

Problem 53. Rikka with Defensive Line

Input file:	*standard input*
Output file:	*standard output*
Time limit:	15 seconds
Memory limit:	256 mebibytes

As an undergraduate student at Peking University, Rikka is always busy with her study. Therefore, to relax in the summer vacation, Rikka spends a lot of time playing computer games.

Today, Rikka is playing a game about a war between two tribes. As the player, Rikka controls the warriors of one tribe to attack the other tribe. The hostile tribe has n strongholds, and the coordinate of the ith stronghold is (x_i, y_i). A defensive line is built among the strongholds: it is a straight line on the coordinate system which divides the whole plane into two regions. The first goal of the game is to capture this defensive line.

As a sophisticated player, Rikka finds that if some of the strongholds are destroyed and all of the remaining strongholds are strictly lying on the same side of the defense line (lying exactly on the defensive line is not allowed), she can capture the defense line with a very small price. Therefore, Rikka will firstly achieve this condition by destroying the minimum number of strongholds at the beginning of the game.

After repeating the game 10^{100} times, Rikka finds that she can always achieve the goal by destroying at most K strongholds. She is curious about this phenomenon. Therefore, Rikka asks Yuta, a master of information technology, to read the source code of the game and find out the reason.

Yuta finds that while initializing, the game will always choose two strongholds and make the defensive line be the only straight line passing through them. Then, to decrease the difficulty, the game will check whether the "easy condition" (i.e., the condition found by Rikka) can be achieved by destroying at most K strongholds: If not, the game will repeat the initialize process until the chosen defensive line is valid.

After observing this secret, Rikka gives the map of the games, i.e., the coordinates of the strongholds. For each stronghold x, Rikka wants you to calculate the number of other strongholds y so that line x, y is a valid defensive line.

Input

The first line of the input contains a single integer $T(1 \le T \le 50)$, the number of test cases.

For each test case, the first line contains two positive integers $n, K(1 \le n \le 5 \times 10^4, 1 \le K \le 50)$, the number of strongholds and the secret constant in the source code.

Then n lines follow, each line with two integers $x_i, y_i(|x_i|, |y_i| \le 10^8)$, which represents the coordinates of the strongholds.

The input guarantees that there are no more than 2 test cases with $n > 1000$ and no two strongholds share the same x coordinate or y coordinate, i.e., for any two different strongholds i, j, $x_i \ne x_j$ and $y_i \ne y_j$.

Output

For each test case, output the single line with n integers which represent the answer for each stronghold.

Example

standard input	standard output
3 5 2 1 0 4 1 3 4 0 3 2 2 5 3 1 0 4 1 3 4 0 3 2 2 5 4 1 0 4 1 3 4 0 3 2 2	2 2 2 2 0 2 2 2 2 0 4 4 4 4 4

Note

In the first two test cases, only the $(1, 2), (2, 3), (3, 4), (1, 4)$ are valid defensive lines where (a, b) represents the straight line passing through the ath and bth strongholds.

In the third test case, any pair of strongholds can from a valid defensive line.

Problem 59. Moon

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	256 mebibytes

Let S be a sphere with radius 1 and center $(0, 0, 0)$. Let $a_0, a_1, \ldots, a_n$ be $n + 1$ points on the surface of S. The positions of $a_1, \ldots, a_n$ are fixed while the position of a_0 is a uniform random point on the surface of S. Let f be 1 if there exists a hemisphere of S that contains $a_0, \ldots, a_n$ and 0 otherwise. Calculate the expected value of f.

Input

The first line contains an integer n denoting the number of points ($0 \le n \le 100000$).

The i-th line of the next n lines contains three integers x, y, z denoting the point $a_i = \left(\frac{x}{\sqrt{x^2+y^2+z^2}}, \frac{y}{\sqrt{x^2+y^2+z^2}}, \frac{z}{\sqrt{x^2+y^2+z^2}}\right)$ ($-1000000 \le x, y, z \le 1000000, x^2 + y^2 + z^2 \ne 0$).

It is guaranteed that $a_1, \ldots, a_n$ are distinct.

Output

Output the answer.

The answer will be considered correct if its absolute or relative error doesn't exceed 10^{-6}.

Example

standard input	standard output
3 1 0 0 0 1 0 0 0 1	0.875000000000

Problem 61. All Pair Maximum Flow

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	256 mebibytes

You are given an undirected graph. You want to compute the maximum flow from each vertex to every other vertex.

The graph is special. You can regard it as a convex polygon with n points (vertices) and some line segments (edges) connecting them. The vertices are labeled from 1 to n in the clockwise order. The line segments can only intersect each other at the vertices.

Each edge has a capacity constraint.

Denote the maximum flow from s to t by $f(s,t)$. Output $\left(\sum_{s=1}^{n}\sum_{t=s+1}^{n} f(s,t)\right) \bmod 998244353$.

Input

The first line contains two integers n and m, representing the number of vertices and edges ($3 \le n \le 200000, n \le m \le 400000$).

Each of the next m lines contains three integers u, v, w denoting the two endpoints of an edge and its capacity ($1 \le u, v \le n, 0 \le w \le 1000000000$).

It is guaranteed there are no multiple edges and self-loops.

It is guaranteed that there is an edge between vertex i and vertex $(i \bmod n) + 1$ for all $i = 1, 2, \ldots, n$.

Output

Output the answer in one line.

Example

standard input	standard output
6 8 1 2 1 2 3 10 3 4 100 4 5 1000 5 6 10000 6 1 100000 1 4 1000000 1 5 10000000	12343461

Problem 67. Honorable Mention

Input file:	*standard input*
Output file:	*standard output*
Time limit:	5 seconds
Memory limit:	256 mebibytes

Ilya Zban has an array $a_1, a_2, \ldots, a_n$. A segment $[l \ldots r]$ of the array is the array $a_l, a_{l+1}, \ldots, a_r$.

Ilya has q ordered triples of the form (l, r, k), where $1 \le l \le r \le n$ and $1 \le k \le r - l + 1$. For each such triple, he asked you to answer the following query: "what is the largest sum of sums of elements of k non-empty non-intersecting subsegments of the segment $[l \ldots r]$?".

Input

The first line of input contains two integers n and q: the number of elements in the array and the number of queries ($1 \le n, q \le 35\,000$).

The second line contains n space-separated integers $a_1, a_2, \ldots, a_n$: the given array ($-35\,000 \le a_i \le 35\,000$).

The next q lines contain queries. Each of them contains three integers l, r, k: the given segment and the number of non-intersecting subsegments on it that you should find ($1 \le l \le r \le n$, $1 \le k \le r - l + 1$).

Output

Output q integers on separate lines: the answers to the queries.

Examples

standard input	standard output
5 5 -1 2 -3 4 -5 1 5 1 1 5 2 1 5 3 1 5 4 1 5 5	4 6 5 2 -3
5 1 7 7 7 7 7 1 5 1	35

Problem 71. Jiry Matchings

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	512 mebibytes

Ruyi Ji has a tree where the vertices are numbered by integers from 1 to n and each edge has a weight.

For each $k \leq (n-1)$, he asked you to find the largest total weight of a matching with k edges if it exists.

Input

The first line of input contains one integer n: the number of vertices in the tree ($2 \leq n \leq 200\,000$).

Each of the next $n-1$ lines contains three integers u_i, v_i, w_i, describing an edge from u_i to v_i with weight w_i in the tree ($1 \leq u_i, v_i \leq n$, $u_i \neq v_i$, $-10^9 \leq w_i \leq 10^9$).

It is guaranteed that the given graph is a tree.

Output

Output $n-1$ integers: the largest weights of the matchings with $1, 2, \ldots, n-1$ edges. If there is no such matching for the current k, print "?" instead.

Examples

standard input	standard output
5 1 2 3 2 3 5 2 4 4 3 5 2	5 6 ? ?
10 2 8 -5 5 10 5 3 4 -5 1 6 5 3 9 5 1 7 -3 4 8 -5 10 8 -5 1 8 -3	5 10 15 10 ? ? ? ? ?
2 1 2 35	35

Note

In the first sample, with $k = 1$ you should take edge $(2, 3)$ with weight 5. And with $k = 2$ you should take two edges, $(2, 4)$ and $(3, 5)$, with total weight 6. There are no matchings with a greater number of edges.

Problem 73. Flaaffy

Input file:	*standard input*
Output file:	*standard output*
Time limit:	15 seconds
Memory limit:	512 megabytes

So you want to see how Pokémon play games, eh? It's a good day for you — Flaaffy, a sheep-like electric Pokémon, just found an electronic Number Guessing Board™ and it wants to have fun with it!

The board is a five-digit electronic display that can show all integers from 0 to 99 999. When Flaaffy turned it on, all five digits were initially set to 0. On its startup, the board chose a secret integer x in the interval $[L, R]$. Flaaffy wants to guess this number. It can use electric shocks to operate the board in two following ways:

- Change a single digit on the display.
- Ask the board if x is smaller, equal or larger than the number shown on the display.

The game ends if Flaaffy can correctly determine what the hidden number is.

However, each operation depletes the amount of electricity stored by Flaaffy. Therefore, it wants to determine the hidden number in the minimum possible number of shocks. Flaaffy already figured out the optimal strategy, can you?

Input

The first line contains a single integer t ($1 \le t \le 50$) — the number of independent testcases in the file. Each of the following t lines describes a single testcase and contains two integers L, R ($1 \le L < R \le 99\,999$).

Output

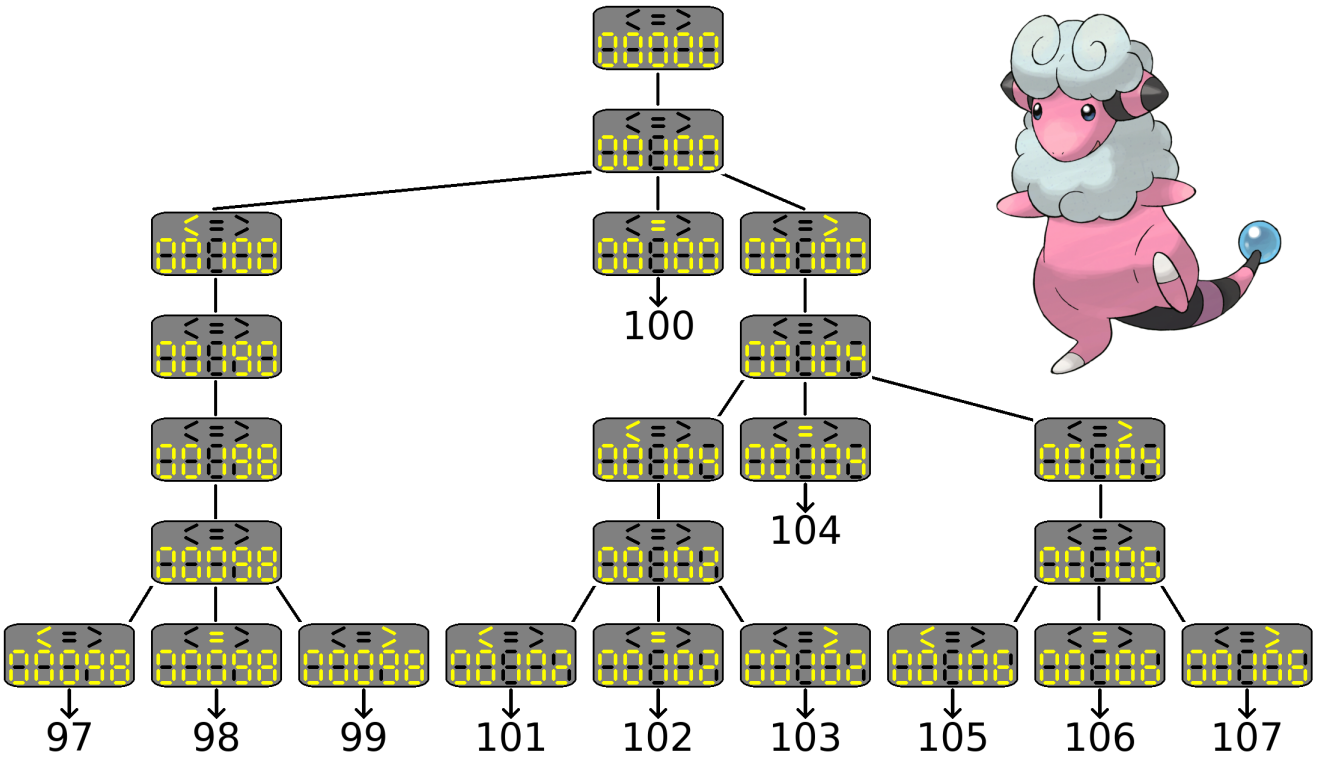
For each testcase, output a single number — the minimum number of shocks Flaaffy needs to produce in order to correctly guess the hidden number.

Example

standard input	standard output
3 97 107 12043 12045 61 69	6 5 7

Note

Here is the decision tree for the first testcase:



Each edge means either changing one digit or comparing x with the number currently on the display. In the leaves of the tree Flaaffy is already sure what the hidden number is.